CROSS-ASSET QUANT RESEARCH

TTF vs. Henry Hub

European vs. US natural gas

Trade: LONG HENRY HUB / SHORT TTF | **Universe:** EU Gas ↔ US Gas | **Horizon:** 1–3M | **Conviction:** MEDIUM

Europe pays 8.3x the US price for gas. TTF is pricing Qatar's outage as if it lasts, and full US export plants cannot respond. As Qatari cargoes return and US capacity grows, the ratio should fall.

SNAPSHOT

Current Gap	**+1.8σ**
Correlation	**+0.18**
Lookback	**9 yrs**
Times This Wide	**3% of wks**
Long Leg	**Henry Hub**
Short Leg	**Dutch TTF**
Target	**+1.0σ**
Stop	**+2.5σ**
R/R	**1.05x**

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Executive Summary

RECOMMENDATION: LONG HH / SHORT TTF | Gap: +1.8σ | Target: +1.0σ | Stop: +2.5σ | Horizon: 1–3M | **MEDIUM CONVICTION**

01

Economic Mechanism

US gas reaches Europe only on LNG ships. When TTF trades far above the landed cost of US LNG (1.15 × Henry Hub + ~$4.50 to liquefy, ship and regasify), exporters buy more US gas and sell it into Europe. That lifts Henry Hub, adds supply in Europe and pulls the two prices together. The link pauses when US export plants run full.

02

Quantified Dislocation

Europe pays 8.31x the US price ($25.24 vs $3.04 per MMBtu). The post-2022 average is 4.70x. That is +1.8σ, wider than 97% of weeks since 2022. TTF sits $17.25 above the landed cost of US gas, against a $9.97 average. TTF is pricing Qatar's outage as lasting while Qatari tankers already resumed Hormuz crossings in September.

03

Trade & Failure Conditions

Long Henry Hub / short TTF futures, 1:1 notional. Entry 8.31x, target 6.49x (+1σ, +25%), stop 10.5x (+2.5σ, −24%), 1–3 months. Converges as Qatar's force majeure lifts (extended through late Q4) and Golden Pass adds US export demand. Breaks if the outage runs through a cold winter with EU storage at 71%.

Relationship & Economic Rationale

Economic Mechanism

US gas reaches Europe only as LNG. When TTF trades above the landed cost of US LNG (1.15 × HH + ~$4.50), exporters buy more Henry Hub gas and ship it. Henry Hub rises, Europe gets more supply, and the ratio falls back.

Transmission Channels

1. Arbitrage: TTF > landed cost → more US cargoes → ratio falls
2. Feedgas: new US trains pull gas → HH rises → ratio falls
3. Capacity cap: US plants full → ratio can stay wide

Why the Link Should Persist

The US became Europe's largest LNG supplier after Russian pipeline flows collapsed in 2022, so the ships keep the markets linked. The link stalls only while US plants are full, which is today: margin $17.25 vs $9.97 average.

Key Numbers

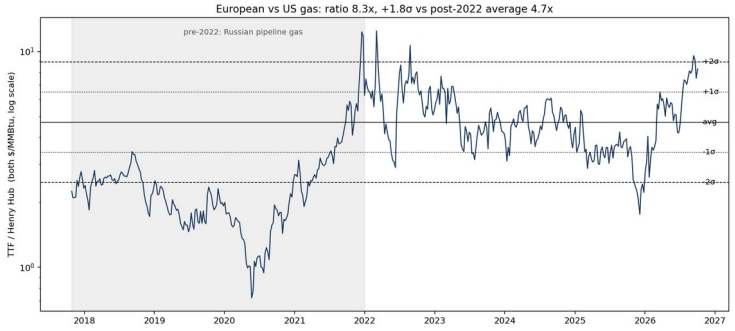
Asset Y	**Henry Hub**
Asset X	**Dutch TTF**
Data Range	**Oct 2017–Oct 2026**
Data Source	**Yahoo (weekly)**
Measure Used	**Ratio TTF ÷ HH**
Correlation (Full)	**+0.18**
Correlation (1Y)	**+0.15**
Avg Gap	**4.70x (log avg)**
Current Gap	**8.31x**
Current Z-Score	**+1.8σ**
Largest Past Gap	**12.45x (+3.0σ)**

Historical Evidence

How We Measured It

Ratio = TTF ÷ Henry Hub, both in $/MMBtu (TTF EUR/MWh × EURUSD ÷ 3.412). Weekly closes, Oct 2017 to Oct 2026, Yahoo Finance. A ratio fits because the landed cost of US LNG scales with Henry Hub. z-scores use the post-2022 average, since losing Russian pipeline gas reset the normal level.

The Chart



Does the Relationship Hold Over Time?

Period	Corr	Simple Avg	Max z	# >2σ	Held?
Full 2017–26	**+0.18**	**3.79x**	**+3.0σ**	**4**	**Reset '22**
Pre-2022	+0.24	2.48x	+3.0σ	1	Y
Post-2022	+0.15	4.95x	+3.0σ	3	Y
Last 3Y	+0.12	4.50x	+2.2σ	1	Y
Last 1Y	+0.15	4.93x	+2.2σ	1	Open
2022 shock	+0.22	6.26x	+3.0σ	2	Y
Today	**—**	**8.31x**	**+1.8σ**	**—**	**Open**

Robust: today's gap is +1.4σ to +2.1σ across six definitions of normal. Post-2022, each 1σ of gap predicted a ~15% fall in the ratio over 13 weeks (p = 0.001). Out of sample (fit to mid-2024, tested after), the direction held ~70% of the time at about half the strength. 3 of 5 trades won with past data only. Not stationary over 2017–26 (p = 0.55), but stationary post-2022 (ADF p = 0.007, half-life ~7 wks).

Current Dislocation — Why Now?

WHY IS THE RELATIONSHIP DISLOCATED NOW?

Size of the Deviation

+1.8σ: only 3% of weeks since 2022 were this wide. At the post-2022 average ratio, TTF would be $14.29 against $25.24 today, or Henry Hub would be $5.37 against $3.04.

What Caused It

TTF repriced and Henry Hub did not. Qatar shipped 18 LNG cargoes in six months against 509 a year earlier, and 17% of its capacity is damaged. US exporters cannot answer because their plants run full.

Which Market Is Right?

Partly both. JKM at $25.70 confirms LNG is scarce today. Our bet is on duration: the ratio prices the outage as lasting, while Qatari tankers resumed Hormuz crossings in September.

Asymmetric Setup

To target (6.49x): +25%. To stop (10.5x): −24%. R/R 1.05x. The edge is the base rate: post-2022, 1σ of gap predicted a ~15% fall over 13 weeks (p = 0.001).

NORMALIZATION CATALYSTS

0–1 Month — **Hormuz LNG transits resume**

Five Qatari carriers crossed Hormuz Sep 19–25, a trickle vs ~85 a month pre-war. Weekly EU storage (71% vs 87% norm).

Expected: gap narrows to ~1.5σ

1–3 Months — **Qatar force majeure + Golden Pass T2**

QatarEnergy force majeure now extended through late Q4. Golden Pass Train 2 (expected late 2026) adds US export demand.

Expected: gap narrows to ~1.0σ

3–6 Months — **Winter passes, US capacity grows**

Golden Pass Train 3 (early 2027). Undamaged Qatari trains back to normal. 17% of Qatar stays offline for ~3 years.

Expected: holds near ~1.0σ, a higher floor than before

Market vs. Normal Relationship

What the Market Says vs. What the Relationship Says

Metric	Today	If Normal	Gap	Why It Should Close
Henry Hub ($/MMBtu)	$3.04	$5.37	−43%	HH needed for a normal ratio if TTF holds. New US export trains pull it up.
Ratio TTF ÷ HH	8.31x (+1.8σ)	4.70x (0.0σ)	+1.8σ	Wider than 97% of weeks since 2022
Story in TTF	Shortage lasts	—	—	Prices Qatar's outage through winter and beyond
Story in Henry Hub	US gas trapped	—	—	Exports maxed, so $17.25/MMBtu of margin goes uncaptured
Correlation	+0.15 (1Y)	+0.18 (full)	−0.03	Stable but low: the link works through capacity, so weekly co-moves are weak

Source of Edge

Why the Gap Exists

A one-off supply shock in Qatar hit the market Europe buys from. The US side cannot respond because export capacity is full.

Why It Hasn't Closed

Arbitrage needs liquefaction plants and ships. Every extra US cargo would earn about $17.25/MMBtu, and there is no spare plant to make it.

What Proves You Right

Qatar lifting force majeure (now extended through late Q4) and Golden Pass Train 2 starting. If both slip past January, we are wrong on timing.

Trade Expression

POSITION PARAMETERS

Structure	**LONG HH / SHORT TTF**
Instruments	**CME NG + ICE TTF futures**
Sizing	**1:1 notional**
Entry	**+1.8σ (8.31x)**
Target	**+1.0σ (+24.7%)**
Risk/Reward	**1.05x**
Stop	**+2.5σ (−23.5%)**
Horizon	**1–3 months**
Exit Plan	**Take profit at +1.0σ (6.49x)**

TRADE RATIONALE

Why This Expression

Short TTF alone is a bet on gas prices. The pair removes the global gas direction and keeps only the Europe vs US gap.

Catalyst

Qatar force majeure decision (extended through late Q4) and Golden Pass Train 2 start (late 2026).

What Convergence Looks Like

Both legs move. TTF eases as Qatari cargoes return. Henry Hub rises as new US trains pull more gas into export.

Cost of Holding

Dec-26 contracts expire late Nov. Roll once into Jan-27 if still open. Costs: margin, the roll, EUR exposure on the TTF leg.

Liquidity

CME Henry Hub is the world's most traded gas contract. ICE Dutch TTF is Europe's benchmark. $1M a leg = ~33 HH and ~16 TTF contracts.

Thesis Invalidation

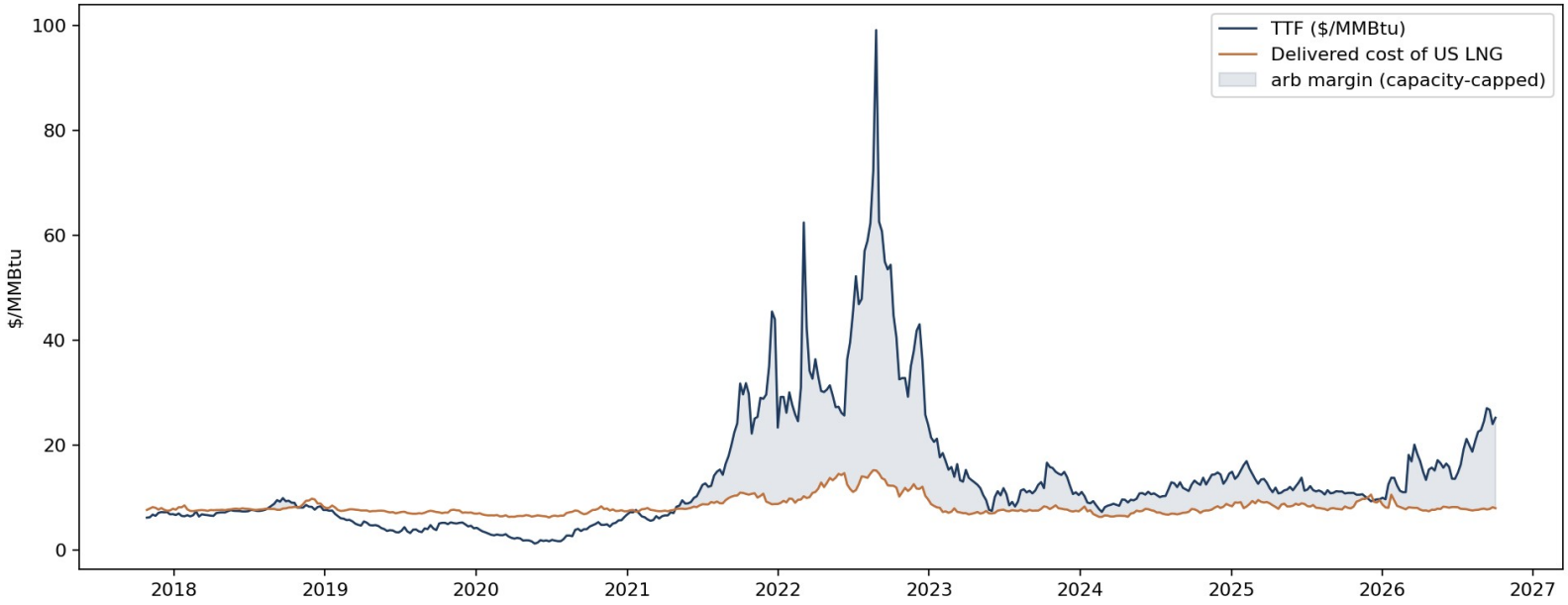
Exit if: (1) ratio hits 10.5x, (2) Qatar extends force majeure past January, (3) Golden Pass Train 2 slips to mid-2027.

Failure Conditions — When Are We Wrong?

Risk	Sev	Prob	Description	Mitigant / Hedge
Permanent Damage	HIGH	MED	17% of Qatar's capacity is offline for ~3 years and North Field expansion slipped to H1 2027. The fair ratio resets above 4.70x.	Target +1σ rather than the average. Exit if Qatar extends force majeure past January.
Winter Squeeze	HIGH	MED	EU storage is 71% full vs an 87% norm going into winter. A European cold snap lifts TTF faster than Henry Hub.	History is mixed: the ratio fell Sep to Dec in 5 of 8 years (avg −7.5%) because US cold lifts HH too. Hard stop at 10.5x.
Hidden Driver	MED	MED	Global LNG scarcity is the real driver. JKM at $25.70 shows Asia pays what Europe pays, so TTF is fair vs Asia.	Track TTF vs JKM. If Asia outbids Europe, US cargoes go east and the gap persists. Exit on that signal.
Hormuz Escalation	HIGH	MED	Renewed US–Iran fighting halts Qatari transits again and TTF spikes, as it did when the war resumed.	Stop at 10.5x. Keep size small enough that a gap through the stop is survivable.
Divergence Widens	MED	MED	With past data only, the 2021 entries lost 50% and 77% while Russian flows were cut. The signal fails in a regime change.	Pre-set stop and a 13-week time stop. No adding as it widens.

Appendix A: The Arbitrage in One Chart

Europe pays $25.24 for gas that US exporters can land for $7.99



Landed cost = 1.15 × Henry Hub + ~$4.50 (liquefaction, shipping, regasification). Before 2021, TTF often sat at or below that line, so Europe was not the obvious destination for US cargoes. Since 2022 the margin stayed above $2 in 90% of weeks: every extra cargo would profit, but US plants are full. Today's margin is $17.25 against a $9.97 post-2022 average.

Appendix B: Robustness Checks

1. Is +1.8σ robust to the choice of normal?

Definition of normal	Avg ratio	z today
Full 2017–26	3.32x	+1.73σ
Post-2022 (used in deck)	4.70x	+1.77σ
Since 2023	4.41x	+2.14σ
Since 2024	4.27x	+2.12σ
Trailing 2 years	4.14x	+1.99σ
Trailing 1 year	4.49x	+1.37σ

2. Does a wide gap predict narrowing?

Horizon	Post-2022: per 1σ	p	All data: p
4 weeks	−4.7%	0.013	0.196
8 weeks	−8.6%	0.007	0.214
13 weeks	−14.9%	0.001	0.294

Future change in log(TTF/HH) regressed on today's z (past data only), HAC errors. Significant since 2022, not on all data (2021 regime break). ADF on the ratio: p 0.55 full sample, 0.007 post-2022 (half-life ~35 trading days), 0.052 since 2023.

3. Trades using past data only (no look-ahead)

Entry	Exit	Weeks	Result	Exit reason
Jan 2021	Feb 2021	3	+37.6%	Target
May 2021	Nov 2021	26	−50.2%	Time limit
Nov 2021	Dec 2021	4	−76.6%	Stop
Dec 2021	Apr 2022	15	+80.6%	Target
Mar 2026	Apr 2026	4	+26.0%	Target
Jul 2026	open	10	−11.8%	Open

Entry at z > +2σ (trailing 2y), exit +1σ, stop +3σ, 26 weeks. Both losses came in the 2021 Russian regime break.

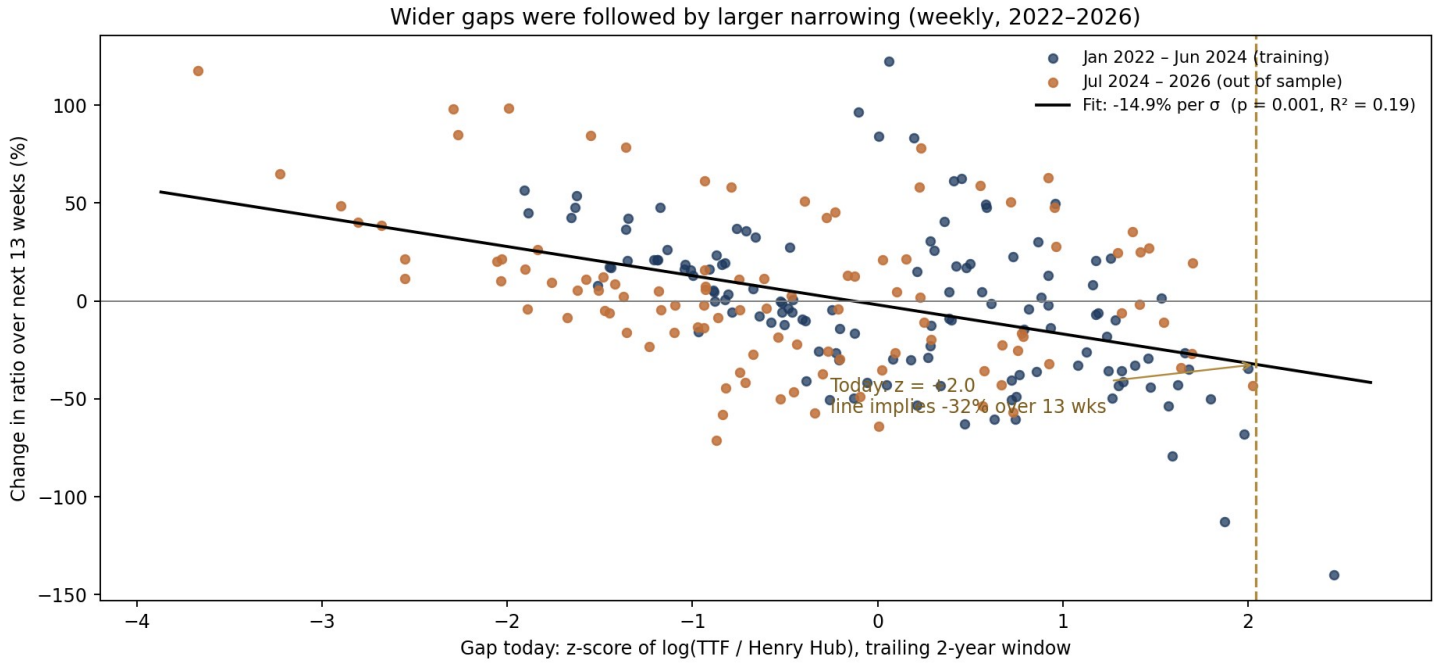
4. Out of sample: fit 2022–Jun 2024, test after

Horizon	Train y	Test y	OOS R^2	Direction right
8 weeks	−12.2%	−5.2%	−0.03	72% of 50 wks
13 weeks	−19.7%	−10.7%	+0.08	69% of 45 wks

The direction held on unseen data. The strength roughly halved and is not significant in the test period alone (p 0.14 to 0.18), so I target only +1σ and keep conviction medium.

Appendix B2: Does a Wide Gap Actually Close?

Each dot is one week since 2022: gap that week vs ratio change over the next 13 weeks



Right side (wide gap): most dots sit below zero, so the ratio usually fell afterward.

Orange dots are out of sample. Same direction, weaker slope (−10.7% vs −19.7% per σ).

The line implies −32% from today. At the out-of-sample strength it is nearer −20%, so the −25% target is plausible, not assured.

Appendix C: How the Pair Was Chosen

~40 pairs screened in Python. Each filter removed pairs for a specific reason.

~40 candidate pairs
rates, FX, metals, energy, equities

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1. Stable link, right sign
out: gold, NOK, CAD, yen, won

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2. Gap stretched today
out: breakevens, GDX/GLD

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3. Gap reverts, not a trend
out: FCX/PAVE, OIH/USO, LITE/FN

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4. Truly cross-market
out: USO/USL (one market)

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5. Wide gap predicts reversal
TTF vs HH: p = 0.001

Pair	Universe	Why it failed	Key number
Gold vs 10y real rate	Commodity ↔ Rates	Link too unstable to define a normal level	Corr −0.52 (22–23) vs −0.13 (24–25)
FCX vs PAVE	Equity ↔ Equity	Spread not stationary. PAVE is mostly rail and steel	Coint p 0.35. ~9% copper users
OIH vs USO	Equity ↔ Commodity	Producers budget off ~$70 deferred oil: the gap was the curve	Coint p 0.18 ex-2020
USO vs USL (front vs 12m oil)	One market	Real curve dislocation, but not cross-asset	Curve z +1.69
NOK vs Brent / EU gas	FX ↔ Commodity	Oil fund keeps revenue abroad. No working link	Corr +0.17 / +0.01
CAD vs WTI (template example)	FX ↔ Commodity	Link broke recently. Tariffs and rates drive CAD	Corr +0.02 (last 3Y)
Yen, won vs Brent	FX ↔ Commodity	Wrong sign or no relationship	Corr ≈ 0
5y breakeven vs Brent	Rates ↔ Commodity	Most stable link in the screen, but not dislocated	z +0.12
GDX vs GLD	Equity ↔ Commodity	Good mechanics, not stretched enough	z +1.36
LITE vs FN	Equity ↔ Equity	Spread trending for two years, so z chased its own mean	z 1.9 on a trend

Data: Yahoo Finance and FRED. Analysis in Python (yfinance, statsmodels). Scripts available on request.